Customer Behavior Analysis

Alfido Tech — Strategic Insights Report

Prepared using E-Commerce Customer Dataset

1. Executive Summary

This report analyzes customer transaction data to identify behavioral segments, purchase patterns, and churn risks for Alfido Tech. Using RFM analysis combined with K-Means clustering, four distinct customer segments were identified: Champions, Loyal, At-Risk, and Hibernating customers.

2. Customer Segment Profiles

Segment	Count	Avg Recency(days)	Avg Frequency	Avg Spend($)
At-Risk	588	152.0	4.1	10104.4
Champions	409	130.2	9.2	26198.1
Hibernating	746	146.8	6.5	17725.4
Loyal	305	586.8	4.3	11718.9

Champions: Bought recently, buy often, spend the most. Reward and retain.

Loyal: Buy regularly with good spend. Strong upsell targets.

At-Risk: Were active but not purchasing recently. Need win-back campaigns.

Hibernating: Last purchase very long ago. Low frequency and spend.

3. Visualizations

Figure 1: Customer Segment Distribution

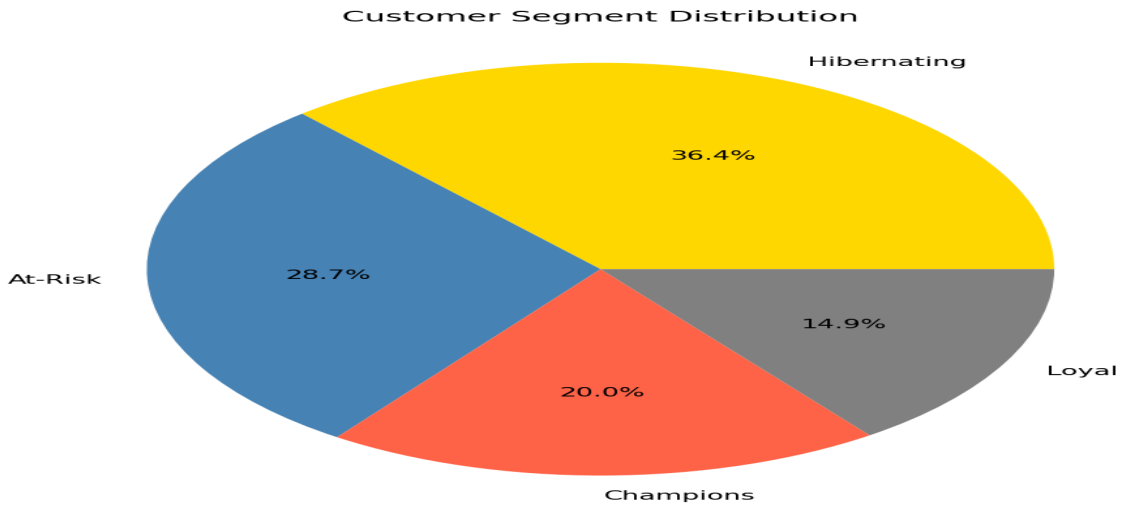


Figure 2: RFM Distributions by Segment

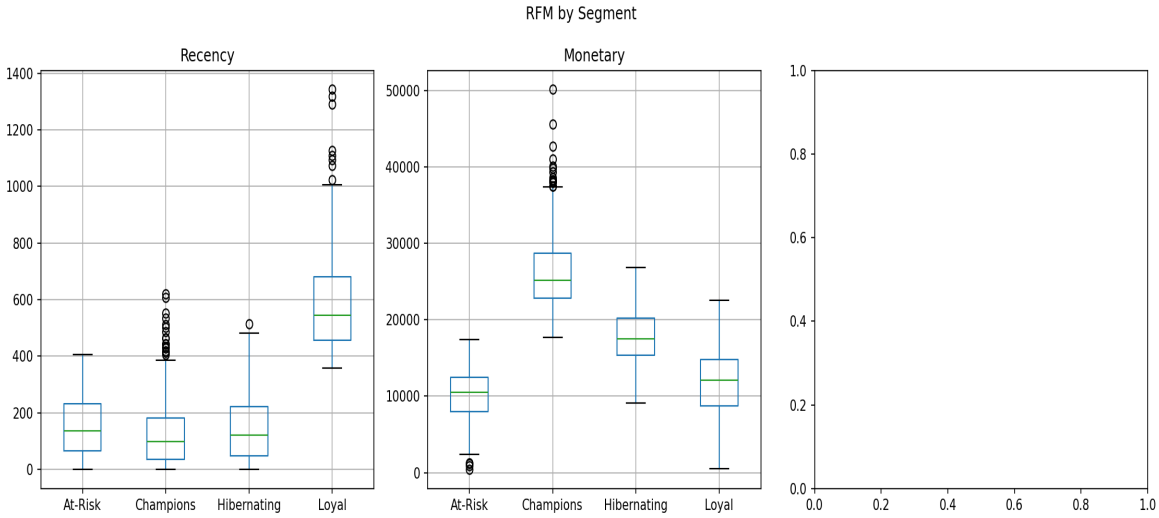


Figure 3: Monthly Revenue Trend

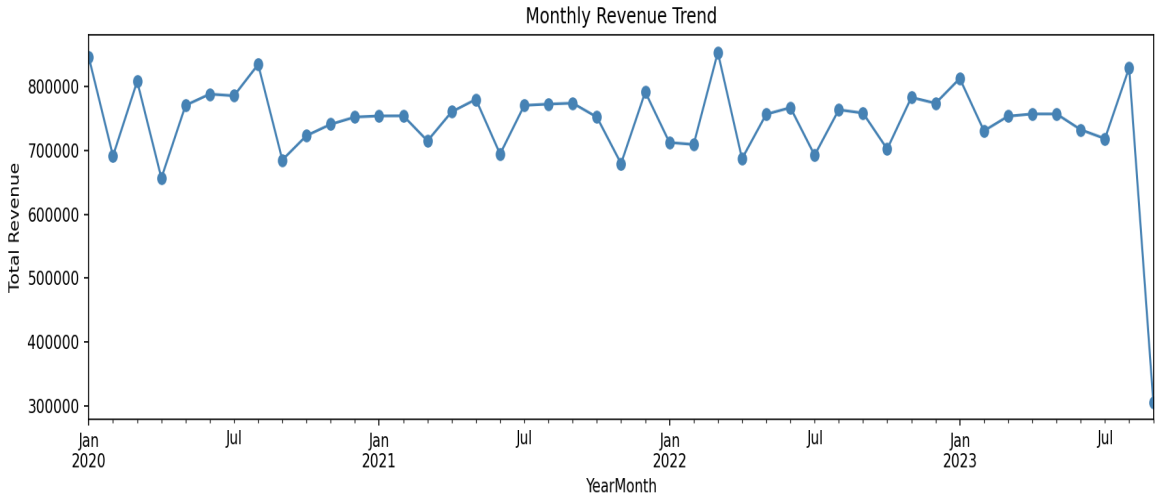


Figure 4: Revenue by Product Category

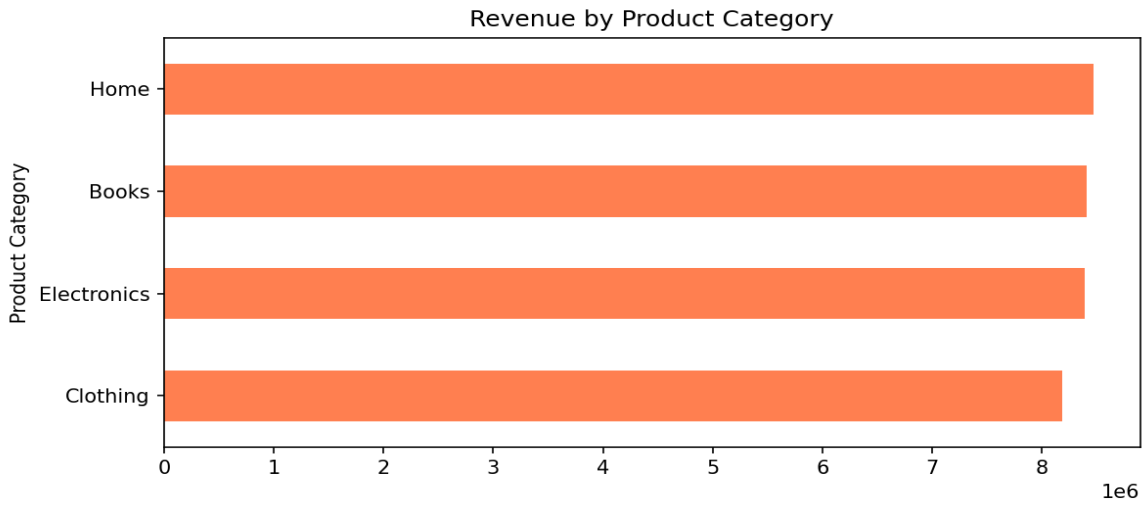


Figure 5: PCA Cluster Visualization

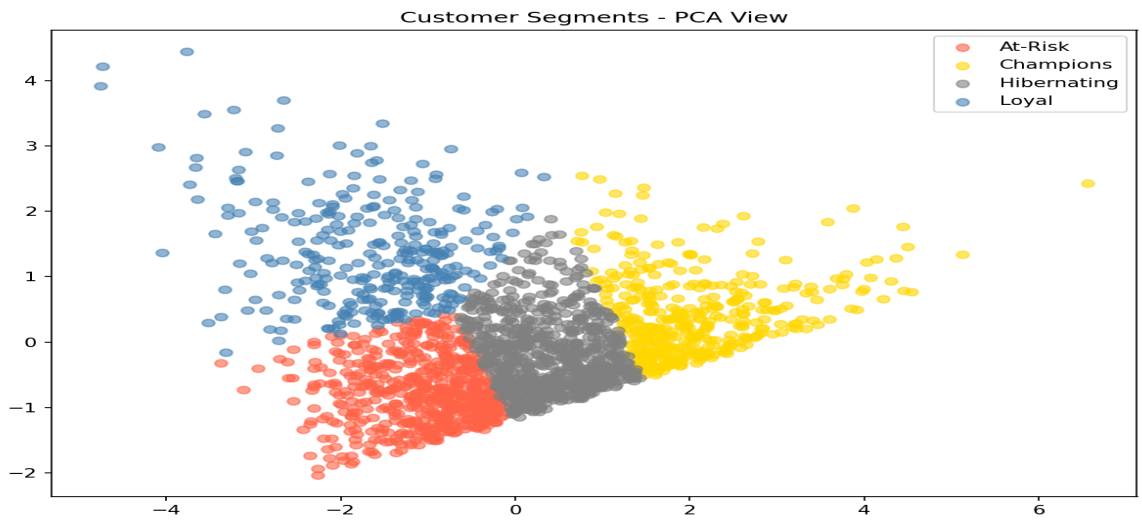
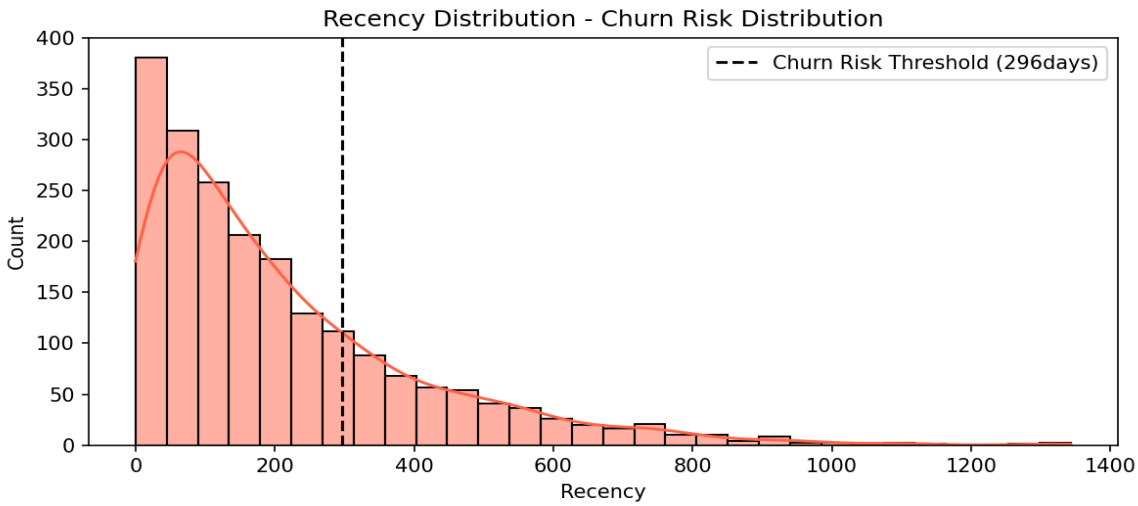


Figure 6: Churn Risk — Recency Distribution



4. Churn Risk Summary

Out of 2048 customers, 431082 (21048.9%) are at high churn risk (recency above 296 days). Immediate re-engagement is recommended.

5. Actionable Recommendations

1. Reward Champions with VIP Programs

Offer exclusive early access, loyalty points, and personalized discounts to top customers.

2. Win-Back At-Risk Customers

Send automated emails with time-limited 10-15% discounts to customers inactive 60+ days.

3. Upsell to Loyal Customers

Cross-sell premium bundles in their favourite categories to increase average order value.

4. Optimize Top Revenue Categories

Increase marketing for Electronics and Clothing. Run promotions on underperforming categories.

5. Implement Real-Time Churn Alerts

Build a live dashboard flagging customers exceeding the 75th percentile recency threshold.